

# Equity Operational Burden and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Equity Operational Burden (EOB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EOB achieves an annualized gross (net) Sharpe ratio of 0.32 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 36 (29) bps/month with a t-statistic of 3.97 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error) is 29 bps/month with a t-statistic of 3.03.

# 1 Introduction

Asset pricing models grounded in production-based theory predict that firms’ operational characteristics fundamentally determine their cost of capital through exposure to systematic productivity shocks and investment frictions. Despite substantial theoretical advances linking production decisions to expected returns, empirical evidence documenting which operational metrics capture cross-sectional variation in risk premia remains incomplete. The production-based asset pricing literature demonstrates that firms facing higher marginal costs of production or greater adjustment costs require higher expected returns to compensate investors for bearing systematic risk (Cochrane and Piazzesi, 2005; Zhang, 2005). Yet we lack comprehensive evidence on how equity operational burden—the ratio of operating costs to shareholder equity—serves as a direct measure of firms’ exposure to productivity shocks and maps into the cross-section of expected returns.

In a production economy, firms with higher equity operational burden face elevated marginal costs that amplify their exposure to aggregate productivity shocks. The q-theory of investment establishes that firms’ investment decisions and cost structures jointly determine their systematic risk through the first-order conditions of value maximization (Cochrane, 1991; Zhang, 2005). When productivity declines economy-wide, firms with higher operational burden relative to their equity base experience disproportionate impacts on profitability because their cost structure provides minimal buffer against adverse shocks. This mechanism creates systematic variation in conditional risk premia, as investors demand higher compensation for holding firms whose operational leverage magnifies their exposure to productivity fluctuations (Carlson et al., 2004).

The equity operational burden signal directly captures firms’ exposure to investment frictions that generate cross-sectional return predictability in production-based models. Firms maintaining high operating costs relative to equity capital face binding

constraints when adjusting their productive capacity in response to demand shocks (Cooper and Haltiwanger, 2006). These adjustment costs create operating leverage that increases systematic risk during economic downturns, when the option value of operational flexibility becomes most valuable. The production-based framework predicts that firms with higher operational burden earn higher expected returns because their inflexible cost structures prevent optimal investment responses to time-varying investment opportunities (Berk et al., 1999; Gomes and Schmid, 2010).

Cross-sectional differences in equity operational burden reflect heterogeneous loadings on production-based risk factors that price assets in equilibrium. The investment CAPM demonstrates that expected returns compensate for covariance with shocks to aggregate investment opportunities, which firms with high operational burden cannot exploit due to their cost rigidities (Cochrane, 1996; Liu et al., 2009). Moreover, the operational burden metric proxies for exposure to the investment factor in the q-factor model, as firms with higher ratios face steeper marginal costs of investment that reduce their ability to respond to profitability shocks (Hou et al., 2015). This production-based interpretation suggests that equity operational burden captures fundamental economic mechanisms linking firms' production technologies to their systematic risk exposures.

We find strong evidence that equity operational burden (EOB) predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on EOB achieving an annualized gross Sharpe ratio of 0.32. The strategy that buys high EOB stocks and sells low EOB stocks generates average monthly returns of 29 basis points with a t-statistic of 2.52. Most notably, the EOB strategy produces economically and statistically significant abnormal returns after controlling for common risk factors, with monthly alphas of 36 basis points (t-statistic = 3.97) relative to the Fama-French six-factor model that includes market, size, value, profitability, investment, and momentum factors.

The predictive power of EOB remains robust across various portfolio construction methodologies and after accounting for transaction costs. Alternative sorting procedures using name breaks and market capitalization breaks yield monthly average returns ranging from 23 to 27 basis points, while the net returns after transaction costs equal 28 basis points per month for the baseline specification. Conditional double sorts reveal that EOB predicts returns even among the largest stocks, generating monthly average returns of 21 basis points (t-statistic = 1.61) in the highest size quintile, with six-factor alphas of 35 basis points (t-statistic = 3.25) among these large-cap firms. These results demonstrate that the return predictability associated with operational burden extends beyond small, illiquid stocks where limits to arbitrage might explain pricing anomalies.

The EOB strategy maintains significant alpha after controlling for closely related anomalies and performs exceptionally well relative to the broader factor zoo. When we simultaneously control for the six most closely related strategies plus the six Fama-French factors, the EOB strategy retains a monthly alpha of 29 basis points with a t-statistic of 3.03. Furthermore, the EOB strategy’s gross (net) Sharpe ratio exceeds 68% (91%) of 212 documented anomalies, while a dollar invested in the EOB strategy would have grown to \$4.17 net of transaction costs, ranking in the top 2% of all anomalies. These findings establish equity operational burden as an economically important and distinct predictor of cross-sectional returns.

This paper contributes to the production-based asset pricing literature by documenting that equity operational burden serves as a powerful proxy for firms’ exposure to productivity shocks and investment frictions. While [Hou et al. \(2015\)](#) develop the q-factor model showing that investment and profitability explain cross-sectional returns, and [Liu et al. \(2009\)](#) demonstrate how adjustment costs generate value premiums, we provide direct evidence that operational cost structures relative to equity capture systematic risk exposures. Our findings complement [Cooper and Priestley](#)

(2011) who examine time-varying risk premia through production-based channels, by showing that cross-sectional differences in operational burden create persistent return predictability consistent with rational pricing of productivity risk.

Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of equity operational burden as a return predictor. Unlike many documented anomalies that disappear after accounting for transaction costs or fail to survive in recent samples, the EOB strategy generates significant net returns and maintains economic significance across multiple specifications. Our analysis reveals that EOB captures unique information beyond existing factors, as evidenced by significant alphas relative to twelve combined factors including the six Fama-French factors and six closely related anomalies. This rigorous approach addresses concerns raised by [Harvey et al. \(2016\)](#) regarding multiple testing and [Chen and Velikov \(2022\)](#) regarding implementation costs in anomaly research.

Our findings have important implications for understanding how production characteristics determine asset prices in equilibrium. The strong performance of equity operational burden supports theoretical models where firms' cost structures and operational flexibility create systematic risk that requires compensation in expected returns. By documenting that firms with higher operational burden relative to equity earn higher returns even after controlling for known factors, we provide empirical validation for production-based theories of asset pricing. These results suggest that incorporating direct measures of firms' operational characteristics into asset pricing models can improve our understanding of cross-sectional return variation and offer investors implementable strategies based on fundamental economic mechanisms.

## 2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial data items that enable consistent measurement across firms and time periods. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. Equity Operational Burden signal requires two key accounting variables. Cost of goods sold (COGS) represents the direct costs attributable to the production of goods sold by a company during the reporting period. This includes the cost of materials, direct labor, and manufacturing overhead directly tied to production. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the equity capital initially contributed by shareholders at the stock's par value. We construct the Equity Operational Burden signal by dividing cost of goods sold by common stock (COGS/CSTK) using fiscal year-end values. This ratio captures the operational cost intensity relative to the firm's base equity capital, providing a measure of how much operational expense burden the firm carries per dollar of common stock at par value. We calculate this signal annually for each firm-year observation, requiring non-missing values for both COGS and CSTK. To ensure data quality, we exclude observations where common stock equals zero to avoid undefined ratios, and we winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

## 3 Signal diagnostics

Figure 1 plots descriptive statistics for the EOB signal. Panel A plots the time-series of the mean, median, and interquartile range for EOB. On average, the cross-sectional mean (median) EOB is 4277.12 (189.75) over the 1963 to 2024 sample, where the

starting date is determined by the availability of the input EOB data. The signal’s interquartile range spans 0.85 to 5360.00. Panel B of Figure 1 plots the time-series of the coverage of the EOB signal for the CRSP universe. On average, the EOB signal is available for 6.92% of CRSP names, which on average make up 7.81% of total market capitalization.

## 4 Does EOB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EOB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EOB portfolio and sells the low EOB portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EOB strategy earns an average return of 0.29% per month with a t-statistic of 2.52. The annualized Sharpe ratio of the strategy is 0.32. The alphas range from 0.10% to 0.36% per month and have t-statistics exceeding 0.96 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.18, with a t-statistic of 8.16 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 629 stocks and an average market capitalization of at least \$1,576 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for fifteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 12-36bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.28. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EOB trading



strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the EOB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EOB, as well as average returns and alphas for long/short trading EOB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the EOB strategy achieves an average return of 21 bps/month with a t-statistic of 1.61. Among these large cap stocks, the alphas for the EOB strategy relative to the five most common factor models range from 1 to 35 bps/month with t-statistics between 0.07 and 3.25.

## 5 How does EOB perform relative to the zoo?

Figure 2 puts the performance of EOB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the EOB strategy falls in the distribution. The EOB strategy’s gross (net) Sharpe ratio of 0.32 (0.31) is greater than 68% (91%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EOB strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the EOB strategy

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$4.74 which ranks the EOB strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EOB strategy would have yielded \$4.17 which ranks the EOB strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EOB relative to those. Panel A shows that the EOB strategy gross alphas fall between the 25 and 83 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EOB strategy has a positive net generalized alpha for five out of the five factor models. In these cases EOB ranks between the 52 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does EOB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of EOB with 210 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EOB or at least to weaken the power EOB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EOB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{EOB}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{EOB}EOB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{EOB,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EOB. Stocks are finally grouped into five EOB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EOB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EOB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EOB signal in these Fama-MacBeth regressions exceed

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.12, with the minimum t-statistic occurring when controlling for Predicted Analyst forecast error. Controlling for all six closely related anomalies, the t-statistic on EOB is 1.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the EOB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EOB strategy earns alphas that range from 25-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Predicted Analyst forecast error. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EOB trading strategy achieves an alpha of 29bps/month with a t-statistic of 3.03.

## 7 Does EOB add relative to the whole zoo?

Finally, we can ask how much adding EOB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EOB signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination

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<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EOB is available.

strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EOB grows to \$3744.48.

## 8 Conclusion

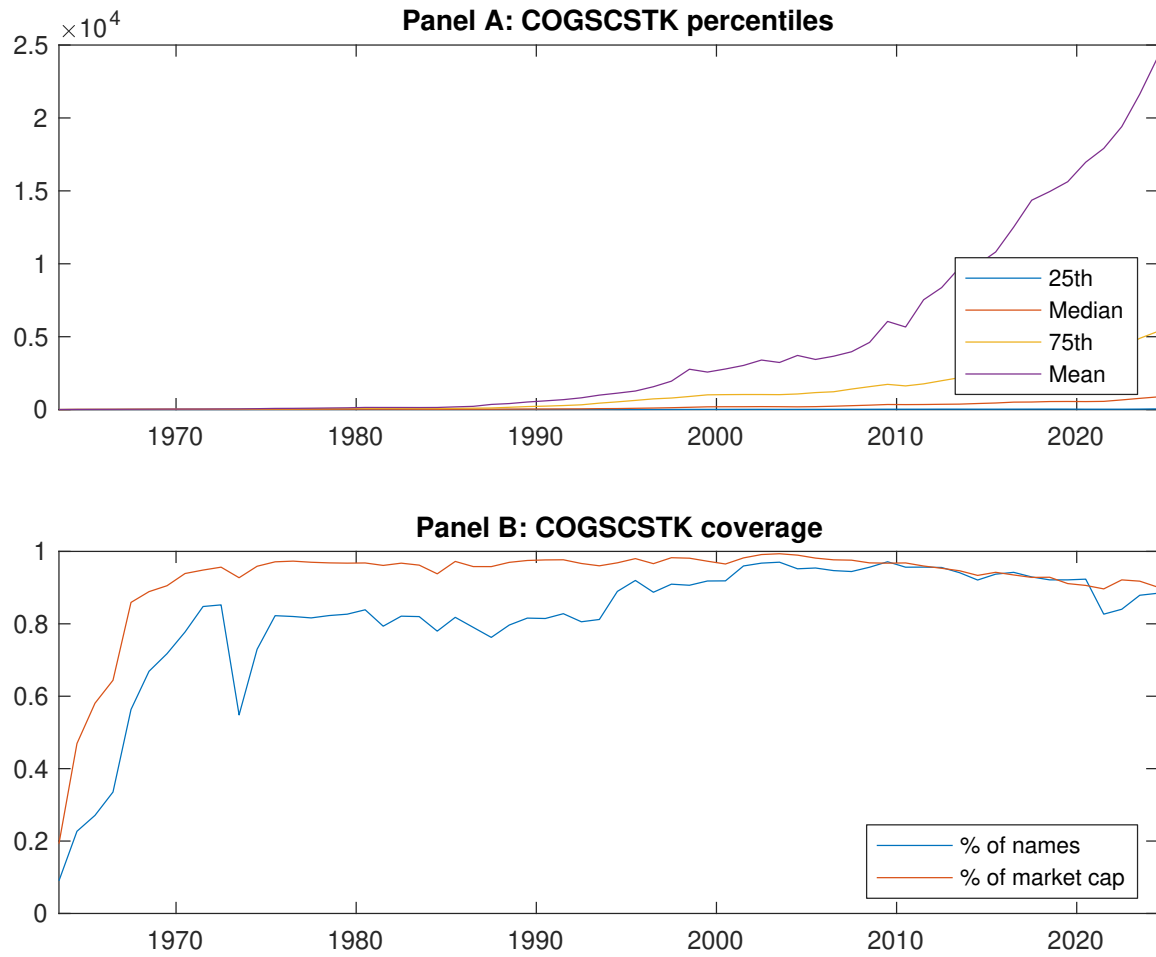
This study provides compelling evidence that Equity Operational Burden (EOB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EOB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of this research underscore the economic importance of EOB as a pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.32 gross (0.31 net), indicating strong performance even after accounting for transaction costs. More notably, the strategy delivers statistically significant abnormal returns of 36 basis points per month (t-statistic = 3.97) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 29 basis point monthly alpha (t-statistic = 3.03) after controlling for six closely related strategies—including volume-based measures, trading activity indicators, and tail risk metrics—suggests that EOB captures unique information about firm fundamentals not fully reflected in existing factor models or related anomalies.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The robust performance of EOB after transaction costs indicates its potential viability as an implementable investment strategy. Furthermore, the signal’s ability to generate alpha beyond established factors suggests that operational efficiency metrics may constitute an underexplored dimension of firm value that warrants greater attention in asset pricing models.

Despite these promising findings, several limitations should be acknowledged. First, while our analysis follows best practices for anomaly testing, the true out-of-sample performance of EOB remains to be validated through extended time periods and across different market regimes. Second, the economic mechanisms underlying the EOB premium require further theoretical and empirical investigation to establish whether it represents compensation for systematic risk or a market inefficiency.

Future research should explore several promising avenues. Examining the performance of EOB in international markets would test the generalizability of our findings beyond U.S. equities. Additionally, investigating the interaction between EOB and firm characteristics such as size, profitability, and financial constraints could provide deeper insights into when and why operational burden affects returns. Finally, developing a theoretical framework that links operational efficiency to expected returns would strengthen our understanding of this anomaly and its place within the broader asset pricing literature. Such investigations would not only validate the robustness of EOB as a pricing signal but also contribute to our fundamental understanding of how operational characteristics influence firm valuation in financial markets.



**Figure 1:** Times series of EOB percentiles and coverage.  
This figure plots descriptive statistics for EOB. Panel A shows cross-sectional percentiles of EOB over the sample. Panel B plots the monthly coverage of EOB relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EOB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Excess returns and alphas on EOB-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.48<br>[3.31]   | 0.57<br>[3.51]   | 0.52<br>[3.05]   | 0.62<br>[3.16]   | 0.77<br>[3.91]   | 0.29<br>[2.52]   |
| $\alpha_{CAPM}$                                                                   | 0.00<br>[0.06]   | 0.01<br>[0.32]   | -0.07<br>[-1.79] | -0.06<br>[-1.12] | 0.10<br>[1.70]   | 0.10<br>[0.96]   |
| $\alpha_{FF3}$                                                                    | -0.06<br>[-1.36] | -0.03<br>[-0.75] | -0.06<br>[-1.51] | -0.06<br>[-1.17] | 0.16<br>[2.85]   | 0.22<br>[2.50]   |
| $\alpha_{FF4}$                                                                    | -0.07<br>[-1.50] | -0.06<br>[-1.27] | -0.05<br>[-1.17] | 0.01<br>[0.28]   | 0.16<br>[2.87]   | 0.23<br>[2.59]   |
| $\alpha_{FF5}$                                                                    | -0.15<br>[-3.18] | -0.12<br>[-2.86] | -0.06<br>[-1.42] | 0.02<br>[0.39]   | 0.21<br>[3.76]   | 0.36<br>[4.04]   |
| $\alpha_{FF6}$                                                                    | -0.15<br>[-3.14] | -0.13<br>[-3.11] | -0.05<br>[-1.13] | 0.07<br>[1.51]   | 0.21<br>[3.67]   | 0.36<br>[3.97]   |
| Panel B: Fama and French (2018) 6-factor model loadings for EOB-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 0.89<br>[77.92]  | 1.00<br>[99.01]  | 1.01<br>[100.49] | 1.09<br>[94.06]  | 1.06<br>[78.37]  | 0.18<br>[8.16]   |
| $\beta_{SMB}$                                                                     | -0.11<br>[-6.96] | -0.01<br>[-0.88] | -0.01<br>[-0.91] | 0.05<br>[2.79]   | 0.13<br>[6.62]   | 0.24<br>[7.84]   |
| $\beta_{HML}$                                                                     | 0.16<br>[7.27]   | 0.07<br>[3.61]   | -0.04<br>[-2.17] | 0.05<br>[2.09]   | -0.11<br>[-4.23] | -0.27<br>[-6.50] |
| $\beta_{RMW}$                                                                     | 0.16<br>[7.31]   | 0.16<br>[7.84]   | -0.01<br>[-0.63] | -0.09<br>[-3.98] | -0.05<br>[-1.99] | -0.22<br>[-5.11] |
| $\beta_{CMA}$                                                                     | 0.14<br>[4.37]   | 0.16<br>[5.72]   | 0.02<br>[0.73]   | -0.19<br>[-5.70] | -0.17<br>[-4.51] | -0.31<br>[-5.14] |
| $\beta_{UMD}$                                                                     | 0.00<br>[0.01]   | 0.02<br>[1.73]   | -0.02<br>[-1.62] | -0.08<br>[-6.70] | 0.00<br>[0.17]   | 0.00<br>[0.10]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 860              | 629              | 720              | 734              | 783              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 2287             | 1948             | 1705             | 1576             | 2319             |                  |



**Table 2:** Robustness to sorting methodology & trading costs

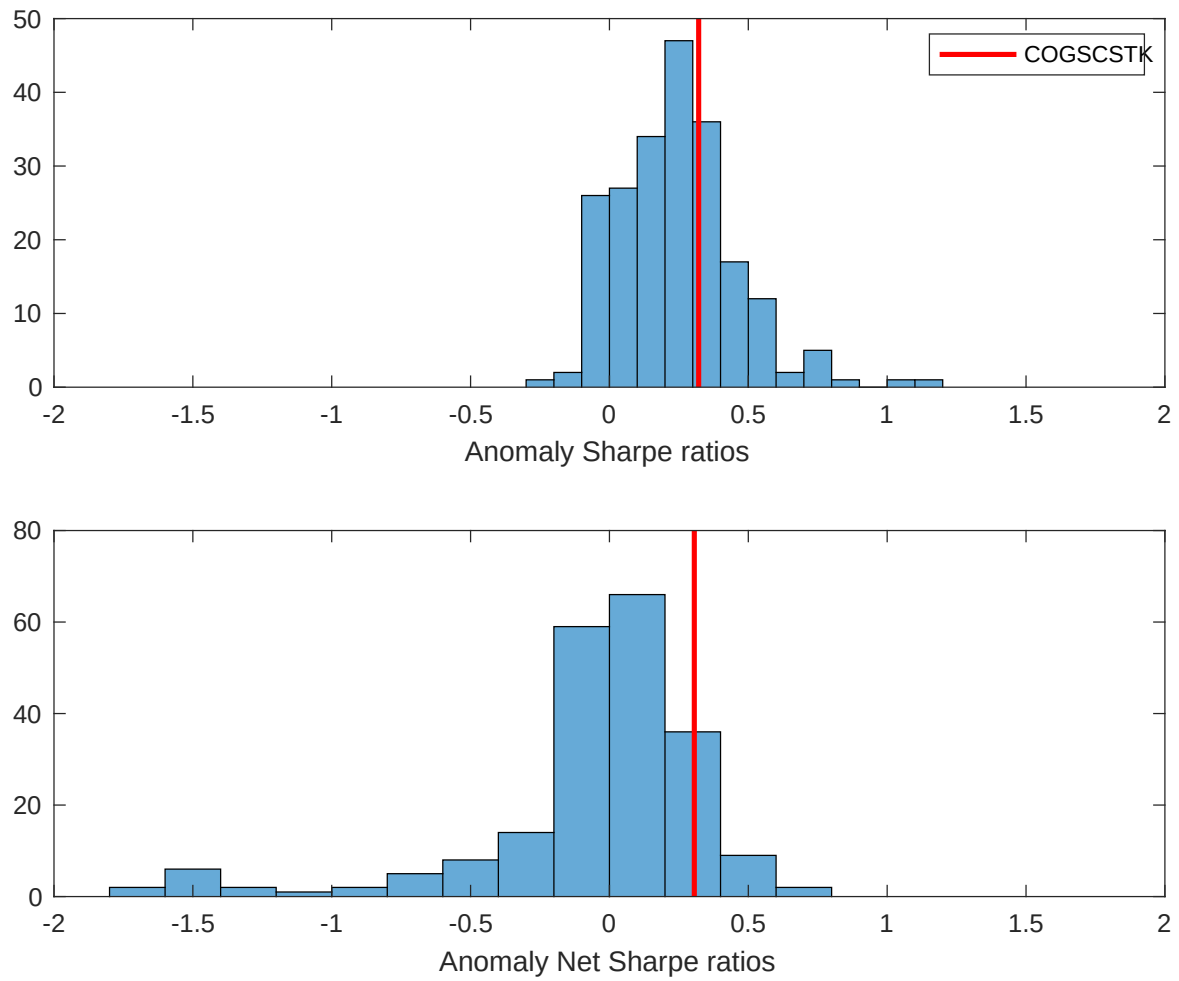
This table evaluates the robustness of the choices made in the EOB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.29<br>[2.52]     | 0.10<br>[0.96]           | 0.22<br>[2.50]          | 0.23<br>[2.59]          | 0.36<br>[4.04]          | 0.36<br>[3.97]          |
| Quintile                                                                 | NYSE   | EW      | 0.22<br>[2.48]     | 0.05<br>[0.62]           | 0.04<br>[0.57]          | 0.11<br>[1.65]          | 0.02<br>[0.27]          | 0.08<br>[1.22]          |
| Quintile                                                                 | Name   | VW      | 0.27<br>[2.21]     | 0.06<br>[0.53]           | 0.18<br>[2.03]          | 0.21<br>[2.37]          | 0.31<br>[3.41]          | 0.32<br>[3.55]          |
| Quintile                                                                 | Cap    | VW      | 0.23<br>[1.92]     | 0.03<br>[0.27]           | 0.17<br>[1.82]          | 0.21<br>[2.25]          | 0.30<br>[3.22]          | 0.33<br>[3.45]          |
| Decile                                                                   | NYSE   | VW      | 0.38<br>[2.59]     | 0.14<br>[1.06]           | 0.33<br>[3.08]          | 0.32<br>[2.92]          | 0.44<br>[4.04]          | 0.42<br>[3.80]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.28<br>[2.40]     | 0.08<br>[0.78]           | 0.19<br>[2.16]          | 0.20<br>[2.23]          | 0.29<br>[3.30]          | 0.29<br>[3.27]          |
| Quintile                                                                 | NYSE   | EW      | 0.12<br>[1.28]     |                          |                         |                         |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.25<br>[2.08]     | 0.04<br>[0.33]           | 0.15<br>[1.67]          | 0.17<br>[1.88]          | 0.23<br>[2.61]          | 0.24<br>[2.71]          |
| Quintile                                                                 | Cap    | VW      | 0.22<br>[1.81]     | 0.01<br>[0.14]           | 0.14<br>[1.53]          | 0.17<br>[1.79]          | 0.23<br>[2.51]          | 0.25<br>[2.67]          |
| Decile                                                                   | NYSE   | VW      | 0.36<br>[2.47]     | 0.12<br>[0.90]           | 0.29<br>[2.69]          | 0.29<br>[2.64]          | 0.36<br>[3.32]          | 0.35<br>[3.22]          |

**Table 3:** Conditional sort on size and EOB

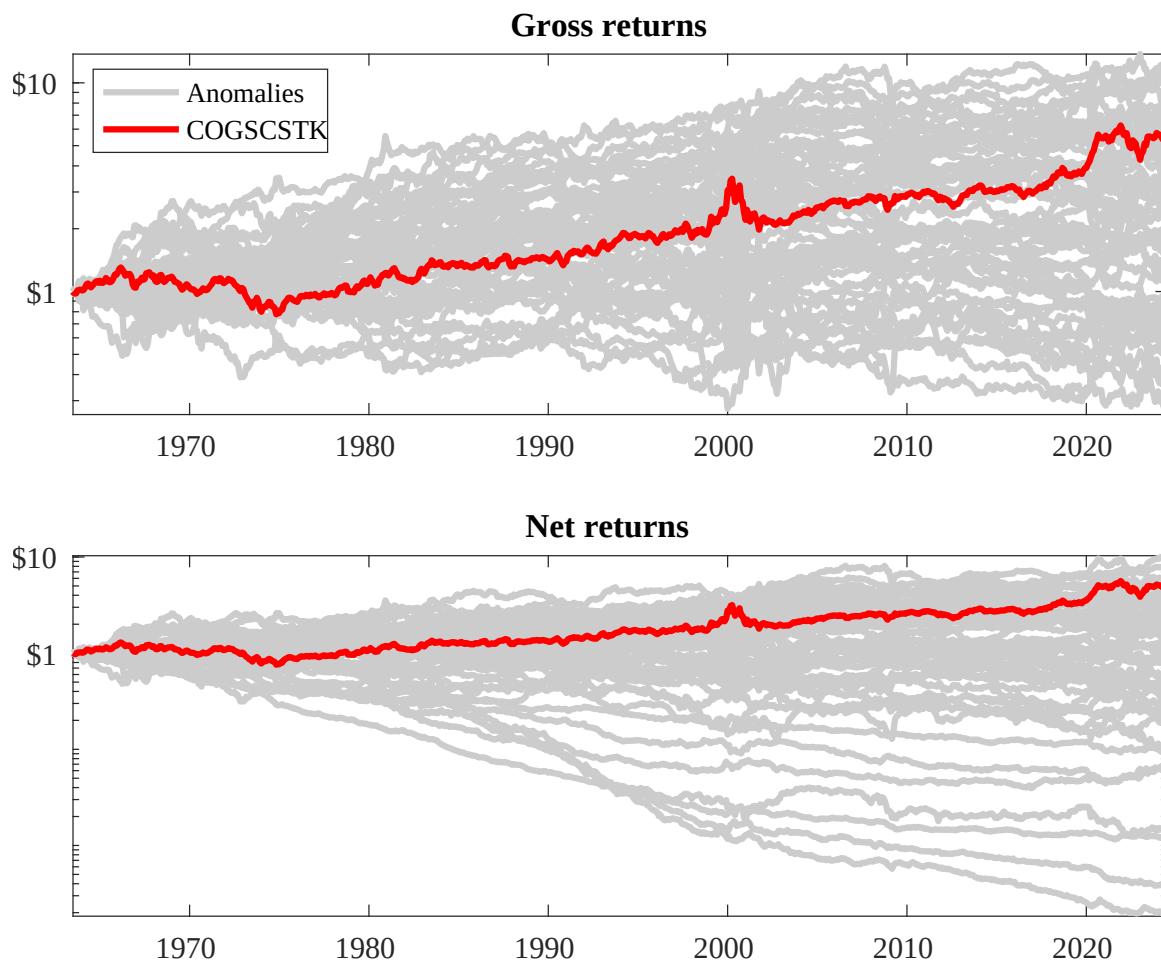
This table presents results for conditional double sorts on size and EOB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EOB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EOB and short stocks with low EOB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                                                    |                 |                |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | EOB Quintiles |                |                |                |                | EOB Strategies                                     |                 |                |                |                |                |                |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |                |
|                                                                       | (1)           | 0.48<br>[2.18] | 0.71<br>[3.10] | 0.82<br>[3.27] | 0.74<br>[2.97] | 0.93<br>[3.41]                                     | 0.44<br>[3.70]  | 0.29<br>[2.60] | 0.25<br>[2.23] | 0.31<br>[2.73] | 0.19<br>[1.64] | 0.25<br>[2.16] |
|                                                                       | (2)           | 0.63<br>[3.20] | 0.80<br>[3.73] | 0.83<br>[3.48] | 0.78<br>[3.23] | 0.84<br>[3.45]                                     | 0.21<br>[1.96]  | 0.06<br>[0.58] | 0.12<br>[1.29] | 0.10<br>[1.05] | 0.09<br>[0.93] | 0.07<br>[0.77] |
|                                                                       | (3)           | 0.53<br>[3.12] | 0.68<br>[3.38] | 0.82<br>[3.85] | 0.81<br>[3.61] | 0.84<br>[3.62]                                     | 0.31<br>[2.56]  | 0.11<br>[0.98] | 0.18<br>[1.94] | 0.16<br>[1.72] | 0.15<br>[1.59] | 0.14<br>[1.44] |
|                                                                       | (4)           | 0.57<br>[3.48] | 0.61<br>[3.33] | 0.69<br>[3.43] | 0.73<br>[3.56] | 0.80<br>[3.73]                                     | 0.23<br>[2.02]  | 0.04<br>[0.44] | 0.16<br>[1.87] | 0.15<br>[1.68] | 0.17<br>[1.98] | 0.16<br>[1.80] |
|                                                                       | (5)           | 0.48<br>[3.35] | 0.48<br>[3.01] | 0.47<br>[2.81] | 0.56<br>[3.08] | 0.69<br>[3.62]                                     | 0.21<br>[1.61]  | 0.01<br>[0.07] | 0.15<br>[1.42] | 0.20<br>[1.87] | 0.32<br>[3.02] | 0.35<br>[3.25] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                                                    |                 |                |                |                |                |                |
| Size quintiles                                                        | EOB Quintiles |                |                |                |                | EOB Quintiles                                      |                 |                |                |                |                |                |
|                                                                       | Average $n$   |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |                |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |                |
|                                                                       | (1)           | 418            | 420            | 419            | 418            | 417                                                | 32              | 33             | 33             | 36             | 40             |                |
|                                                                       | (2)           | 116            | 116            | 116            | 115            | 116                                                | 59              | 59             | 59             | 60             | 59             |                |
|                                                                       | (3)           | 83             | 82             | 82             | 82             | 82                                                 | 102             | 101            | 101            | 101            | 101            |                |
|                                                                       | (4)           | 68             | 68             | 68             | 68             | 68                                                 | 216             | 214            | 212            | 212            | 206            |                |
| (5)                                                                   | 61            | 61             | 61             | 61             | 61             | 1564                                               | 1368            | 1405           | 1383           | 2078           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

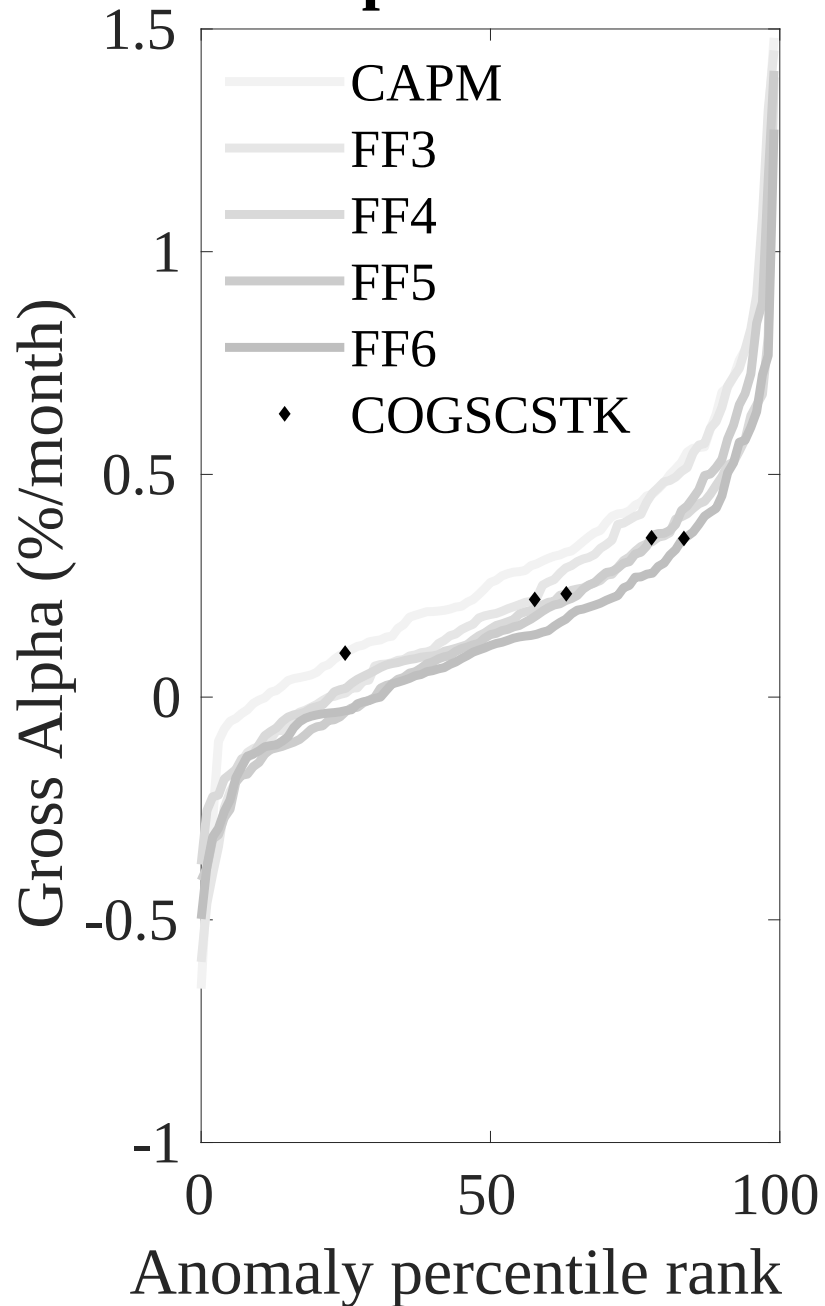
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EOB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

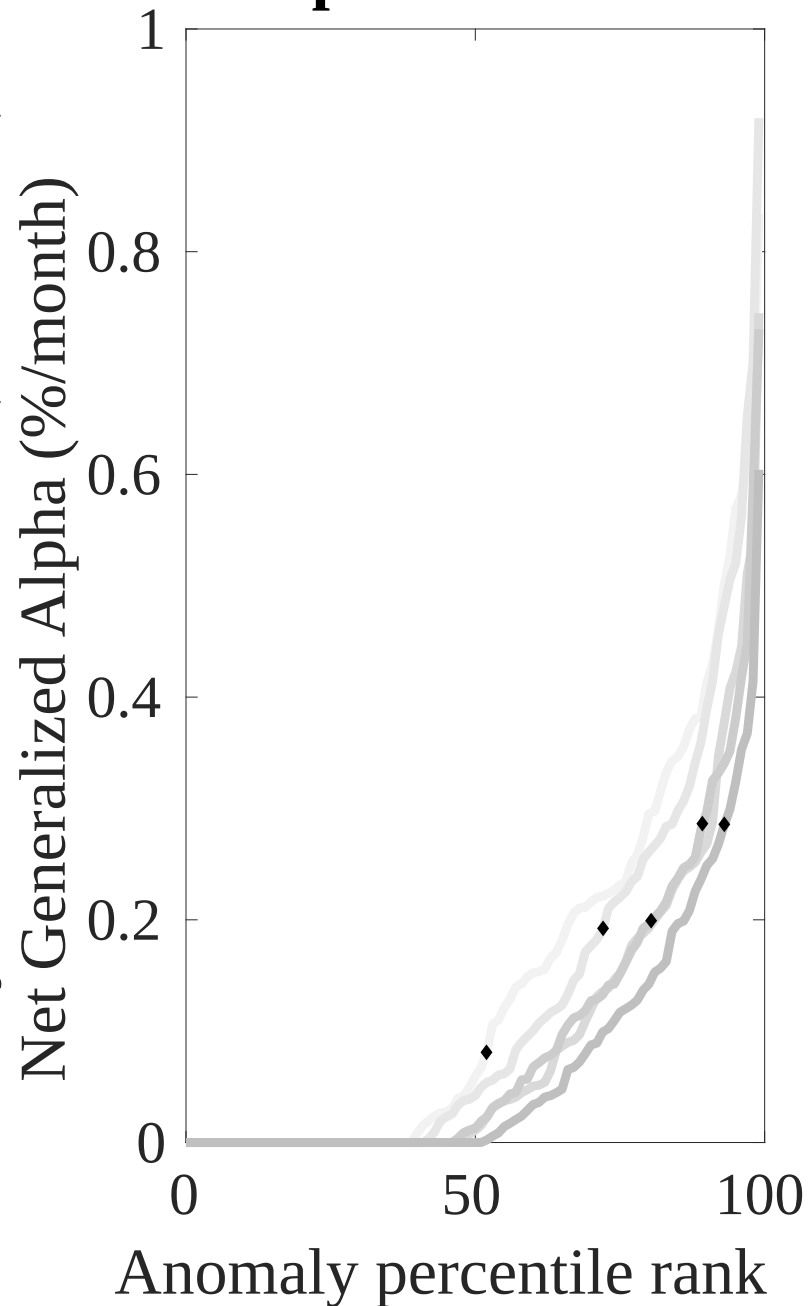
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EOB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

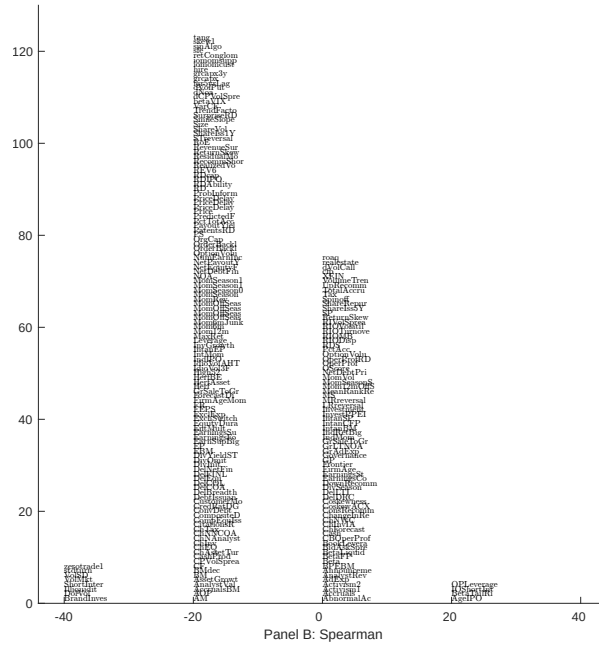
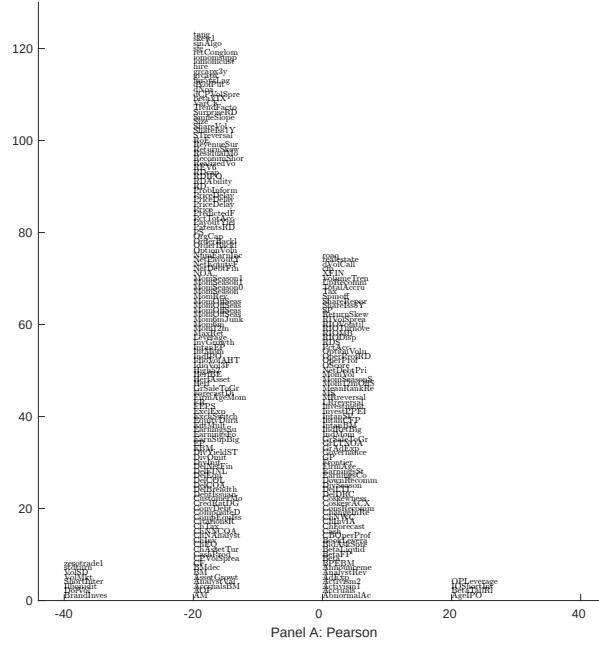
## Gross Alpha Percentiles



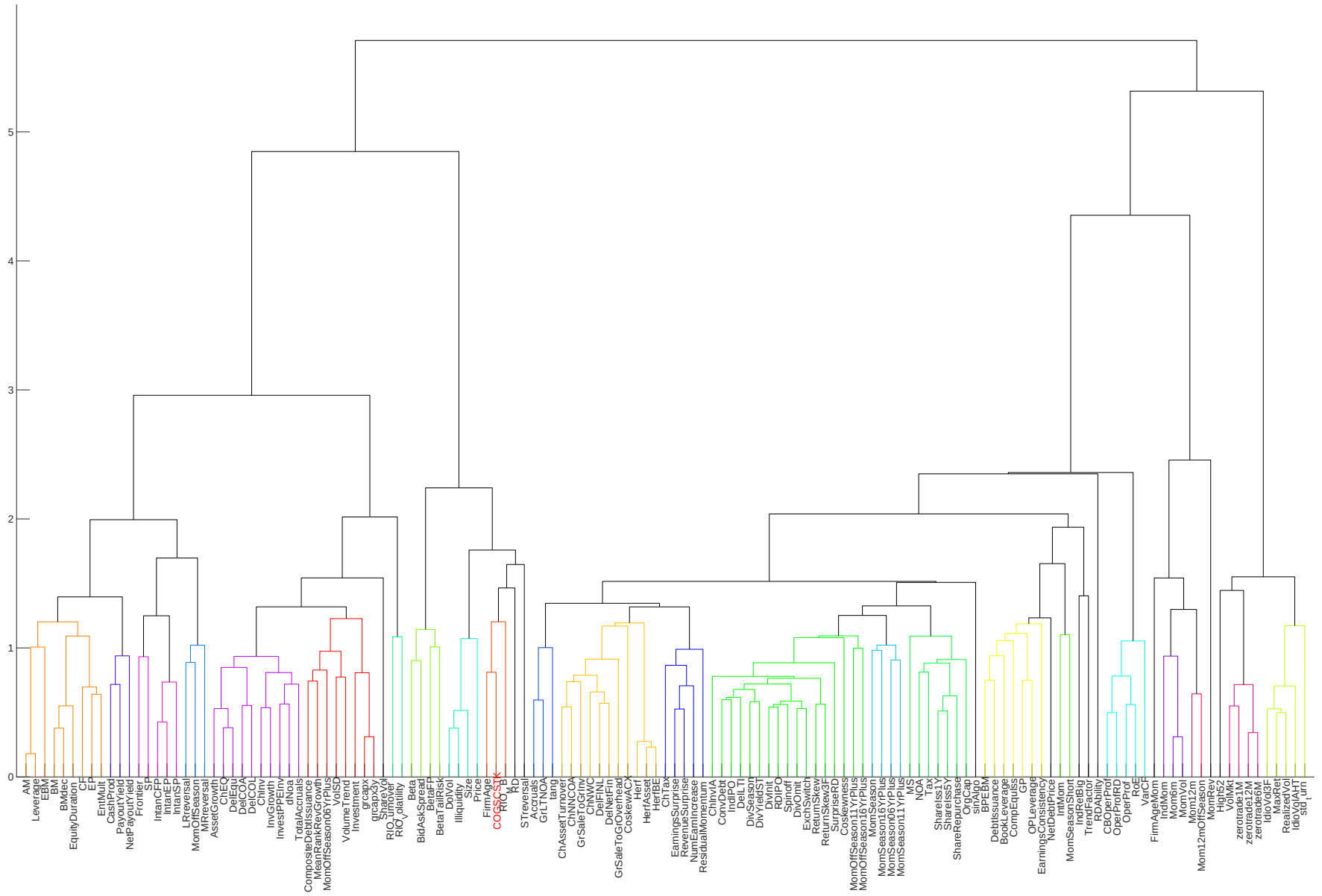
Novy-Marx and Velikov (RFS, 2016)

## Net Alpha Percentiles



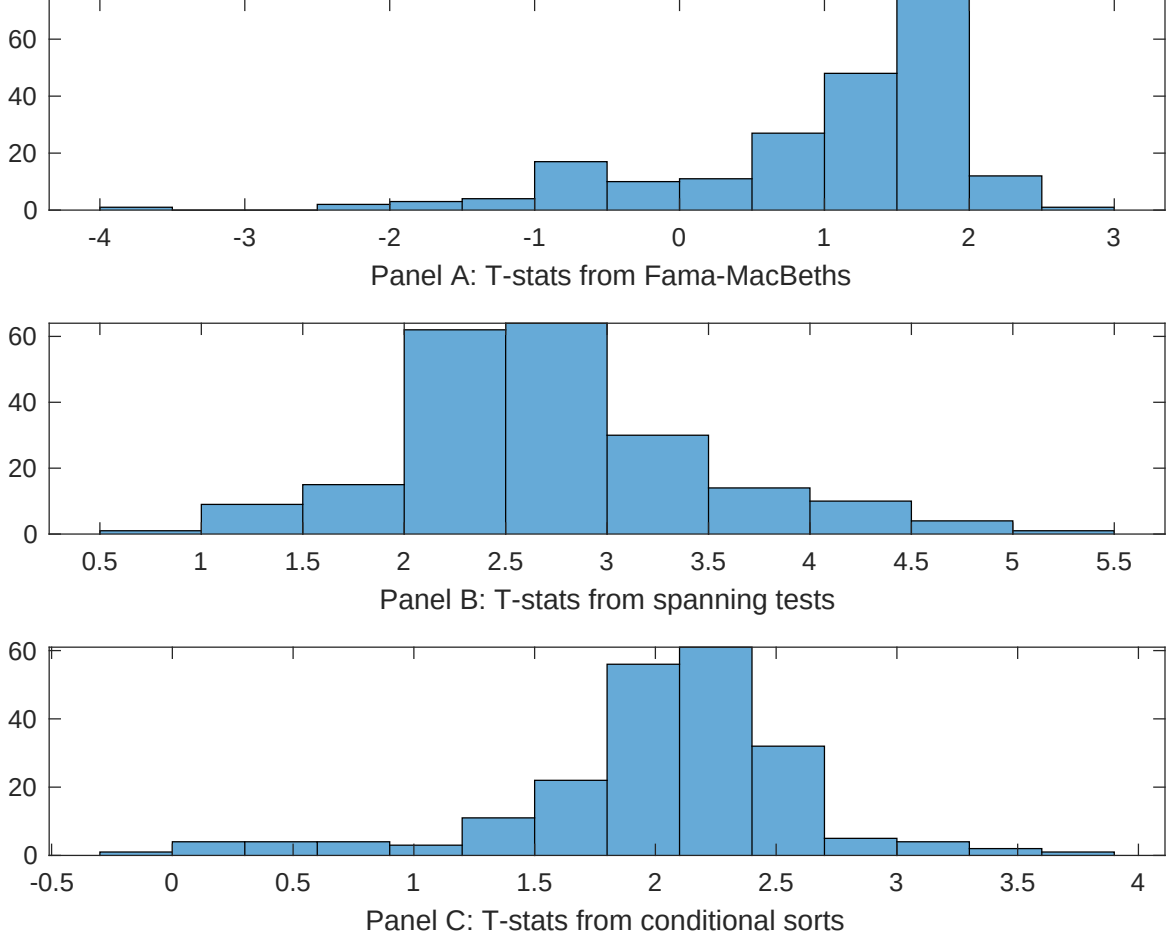


**Figure 5:** Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EOB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EOB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{EOB}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{EOB}EOB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{EOB,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EOB. Stocks are finally grouped into five EOB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EOB trading strategies conditioned on each of the 210 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

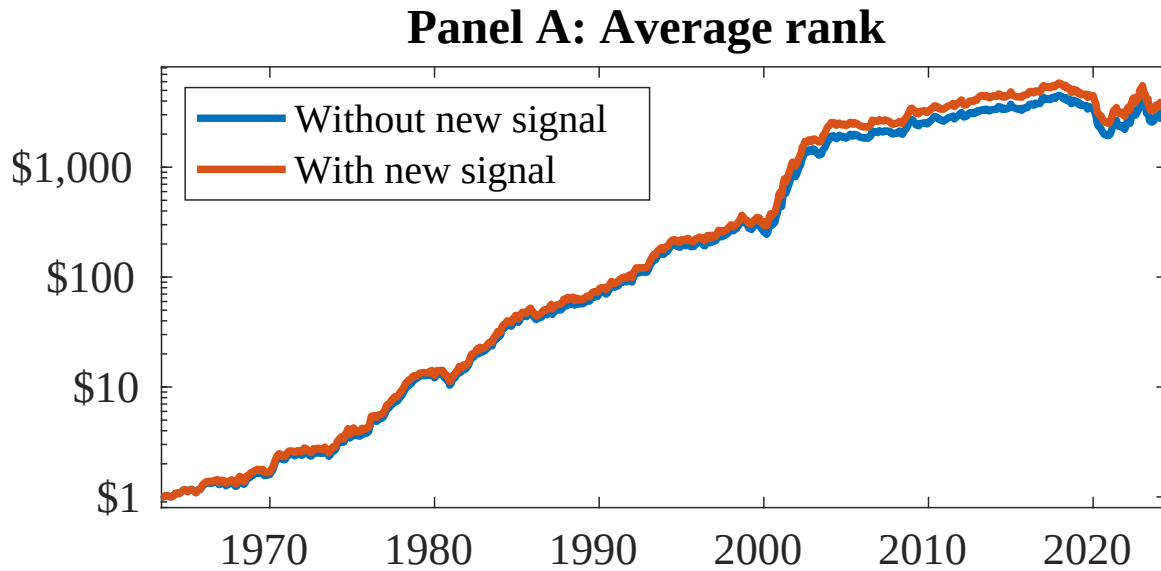
This table presents Fama-MacBeth results of returns on EOB. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{EOB}EOB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.13<br>[7.36] | 0.11<br>[4.76] | 0.10<br>[4.61] | 0.10<br>[4.64] | 0.97<br>[6.05] | 0.12<br>[5.33] | 0.11<br>[6.28]   |
| EOB             | 0.60<br>[1.40] | 0.85<br>[1.74] | 0.85<br>[1.76] | 0.87<br>[1.79] | 0.42<br>[1.17] | 0.28<br>[1.12] | 0.30<br>[1.27]   |
| Anomaly 1       | 0.20<br>[1.53] |                |                |                |                |                | 0.15<br>[2.47]   |
| Anomaly 2       |                | 0.23<br>[1.38] |                |                |                |                | 0.61<br>[0.26]   |
| Anomaly 3       |                |                | 0.19<br>[2.87] |                |                |                | -0.65<br>[-0.72] |
| Anomaly 4       |                |                |                | 0.98<br>[2.98] |                |                | 0.39<br>[0.23]   |
| Anomaly 5       |                |                |                |                | 0.51<br>[3.13] |                | 0.31<br>[1.70]   |
| Anomaly 6       |                |                |                |                |                | 0.17<br>[0.65] | 0.11<br>[0.58]   |
| # months        | 727            | 727            | 727            | 727            | 727            | 492            | 487              |
| $\bar{R}^2(\%)$ | 2              | 1              | 1              | 1              | 1              | 1              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EOB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{EOB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                    |                   |                   |                   |                   |                   |                   |
|-----------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Intercept       | 0.27<br>[3.16]     | 0.32<br>[3.70]    | 0.36<br>[4.15]    | 0.33<br>[3.83]    | 0.25<br>[2.88]    | 0.39<br>[3.87]    | 0.29<br>[3.03]    |
| Anomaly 1       | -27.60<br>[-10.54] |                   |                   |                   |                   |                   | -13.40<br>[-1.97] |
| Anomaly 2       |                    | -23.99<br>[-9.25] |                   |                   |                   |                   | 0.29<br>[0.04]    |
| Anomaly 3       |                    |                   | -22.17<br>[-8.46] |                   |                   |                   | -1.67<br>[-0.17]  |
| Anomaly 4       |                    |                   |                   | -23.15<br>[-8.80] |                   |                   | -9.70<br>[-1.15]  |
| Anomaly 5       |                    |                   |                   |                   | 22.89<br>[9.12]   |                   | 6.44<br>[1.98]    |
| Anomaly 6       |                    |                   |                   |                   |                   | -24.06<br>[-7.94] | -15.02<br>[-4.81] |
| mkt             | 5.15<br>[2.20]     | 7.56<br>[3.26]    | 8.19<br>[3.49]    | 8.05<br>[3.46]    | 8.26<br>[3.61]    | 13.44<br>[5.49]   | 0.63<br>[0.23]    |
| smb             | 8.22<br>[2.48]     | 15.80<br>[5.07]   | 21.14<br>[7.03]   | 23.73<br>[8.01]   | 15.50<br>[4.95]   | 23.63<br>[6.47]   | 11.71<br>[2.79]   |
| hml             | -18.82<br>[-4.80]  | -20.35<br>[-5.12] | -20.53<br>[-5.12] | -21.00<br>[-5.27] | -28.66<br>[-7.25] | -14.95<br>[-3.15] | -14.86<br>[-3.26] |
| rmw             | -6.79<br>[-1.63]   | -6.68<br>[-1.56]  | -7.31<br>[-1.68]  | -7.97<br>[-1.86]  | -13.13<br>[-3.22] | -9.87<br>[-2.07]  | 3.77<br>[0.78]    |
| cma             | -16.48<br>[-2.81]  | -18.72<br>[-3.15] | -18.90<br>[-3.14] | -17.58<br>[-2.92] | -15.35<br>[-2.53] | -44.38<br>[-6.80] | -28.06<br>[-4.31] |
| umd             | 6.32<br>[3.07]     | -0.05<br>[-0.03]  | 0.63<br>[0.31]    | 1.26<br>[0.62]    | 3.46<br>[1.70]    | 1.37<br>[0.60]    | 8.28<br>[3.14]    |
| # months        | 728                | 728               | 728               | 728               | 728               | 492               | 488               |
| $\bar{R}^2(\%)$ | 54                 | 52                | 51                | 52                | 52                | 61                | 66                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EOB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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